AI-Powered Resume Screening & Candidate Ranking System

Internship Project Report

1. Introduction

Recruitment is a critical process for organizations seeking qualified candidates. Modern organizations receive a large number of applications for each job opening, making manual resume screening both time-consuming and inefficient.

To address this challenge, an AI-powered Resume Screening and Candidate Ranking System was developed. The system automates resume evaluation by extracting relevant information, comparing candidate profiles against job descriptions, and ranking applicants according to their suitability.

The project combines Natural Language Processing (NLP), Machine Learning techniques, and automated decision logic to simulate the core functionality of modern Applicant Tracking Systems (ATS).

2. Problem Statement

Recruiters often spend significant time reviewing resumes manually. This process can introduce inconsistencies, delays, and subjective bias.

The objective of this project is to develop an automated screening system capable of:

- Reading resumes from multiple formats
- Extracting relevant information
- Understanding job requirements
- Matching candidate profiles against job descriptions
- Ranking candidates automatically

3. Objectives

The primary objectives of this project are:

- Build an end-to-end NLP pipeline
- Extract structured information from unstructured resumes

- Calculate resume-job similarity
- Generate candidate relevance scores
- Rank applicants automatically
- Reduce recruiter effort

4. Dataset Description

The project utilizes a publicly available Resume Dataset obtained from Kaggle.

Dataset Details:

- Total Records: 586
- Format: CSV
- Main Columns:
  - ID
  - Resume_str
  - Resume_html
  - Category

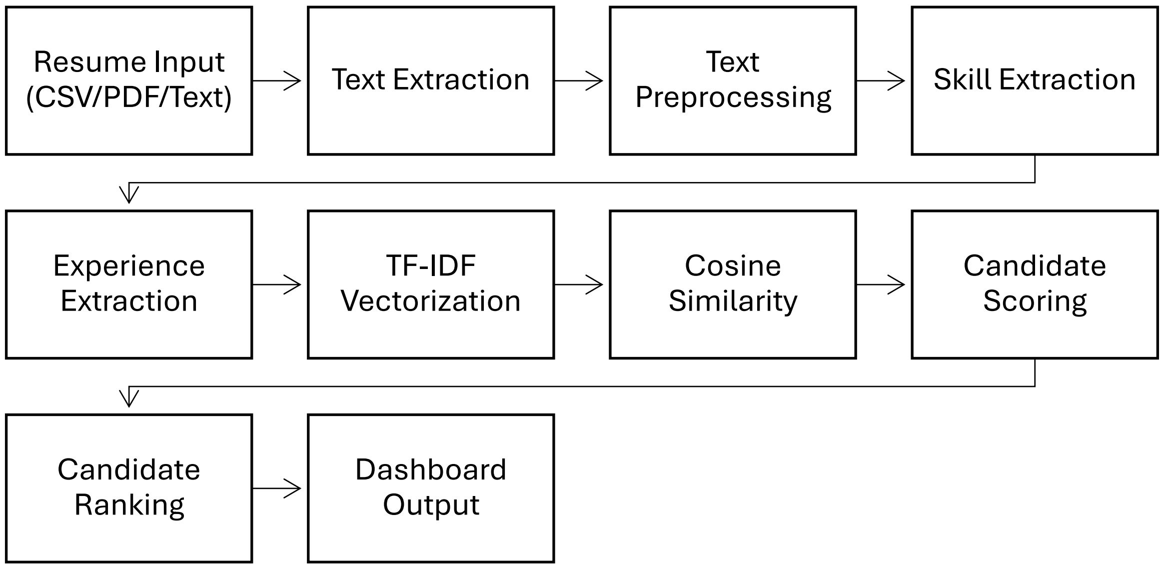
Categories include:

- Information Technology
- Engineering
- Finance
- Banking
- Consultant
- Agriculture
- Healthcare
- Sales
- HR
- Public Relations

In addition to the CSV dataset, PDF resumes and manually entered resume text were used to validate the system.

5. System Architecture

The overall workflow of the system is illustrated below:



6. Methodology

6.1 Resume Ingestion

The system accepts resumes through three different input methods:

- CSV dataset upload
- PDF resume upload
- Direct text input

This design improves flexibility and supports multiple real-world use cases.

6.2 Text Preprocessing

The extracted resume text undergoes preprocessing steps:

- Conversion to lowercase
- Removal of special characters
- Stopword removal
- Lemmatization

These steps improve text quality and prepare the data for feature extraction.

6.3 Skill Extraction

A predefined technical skill dictionary is used to identify important skills within resumes and job descriptions.

Examples include:

- Python
- SQL
- Machine Learning
- Deep Learning
- TensorFlow
- NLP
- Git
- Docker

6.4 Experience Extraction

Regular expression-based pattern matching is used to identify years of experience mentioned in resumes.

6.5 Feature Engineering

Three major features are generated:

1. Skill Match Score
2. Experience Score
3. Similarity Score

These features collectively determine candidate relevance.

6.6 Similarity Scoring

TF-IDF Vectorization converts text into numerical vectors.

Cosine Similarity is then used to measure similarity between candidate resumes and job descriptions.

6.7 Candidate Ranking

Final candidate scores are calculated using weighted scoring.

Candidates are sorted in descending order according to their final scores.

7. Streamlit Dashboard

A Streamlit-based dashboard was developed to provide an interactive user interface.

Features include:

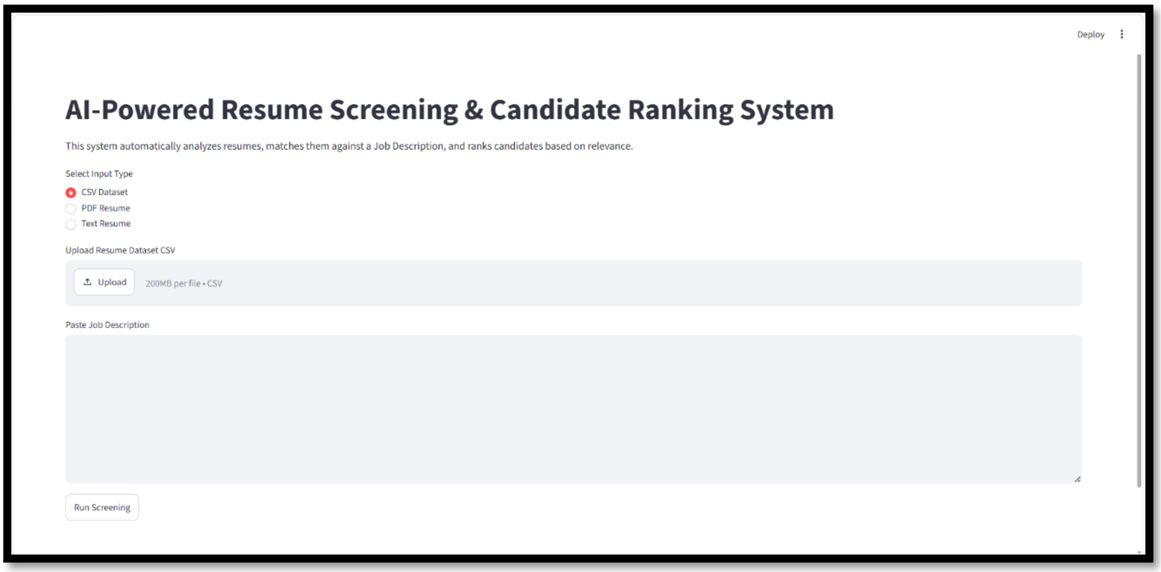
- Resume Upload
- Job Description Input
- Candidate Ranking
- Skill Gap Analysis
- Downloadable Results

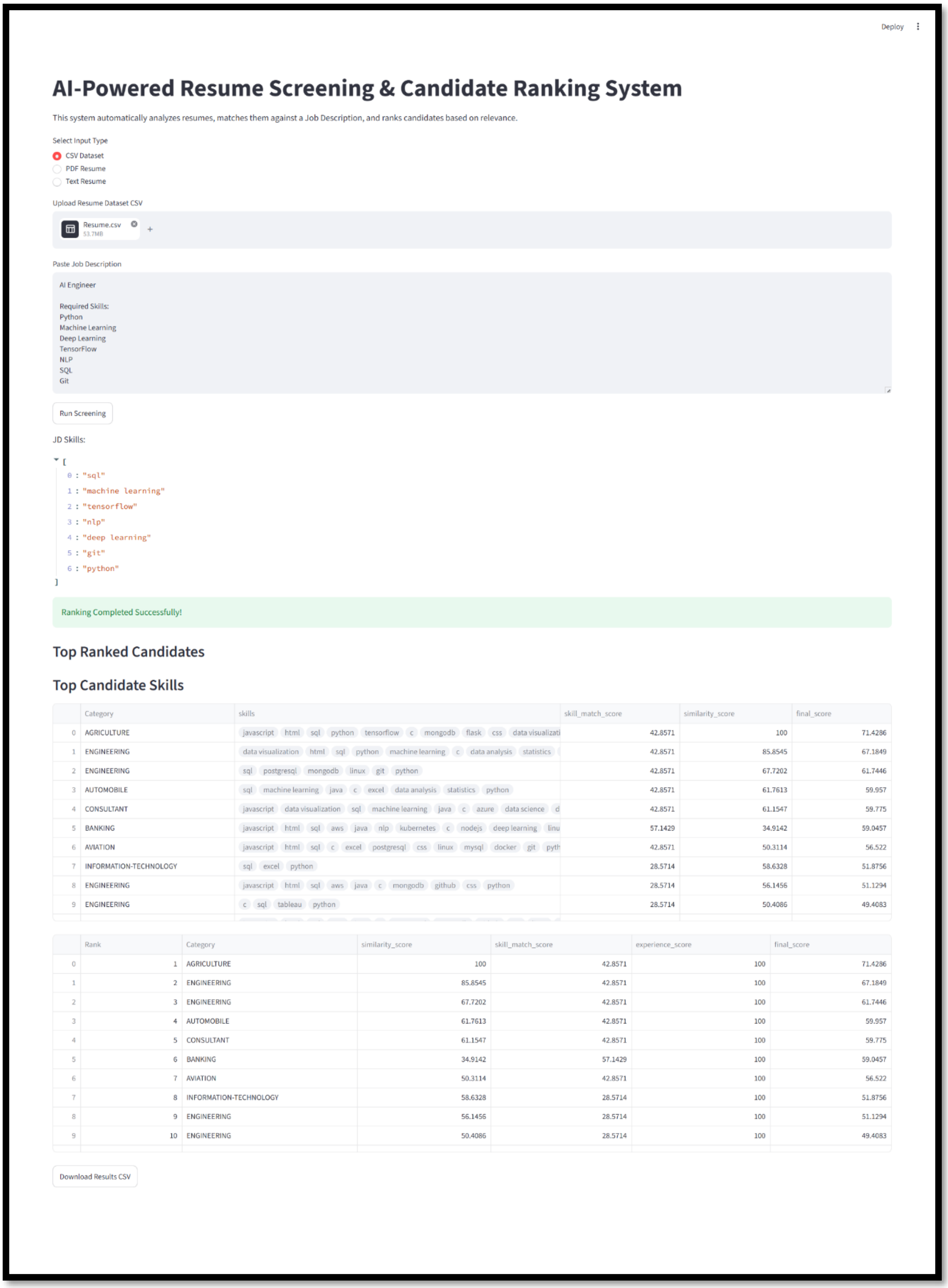
The dashboard enables recruiters to evaluate resumes without interacting directly with the codebase.

8. Results

The developed system successfully:

- Parsed resumes from multiple formats
- Extracted skills and experience
- Calculated resume-job similarity
- Generated candidate rankings
- Identified missing skills





Deploy

AI-Powered Resume Screening & Candidate Ranking System

This system automatically analyzes resumes, matches them against a Job Description, and ranks candidates based on relevance.

Select Input Type

- CSV Dataset
- PDF Resume
- Text Resume

Upload Resume Dataset CSV

Resume.csv 53.7MB

Paste Job Description

AI Engineer

Required Skills:
Python
Machine Learning
Deep Learning
TensorFlow
NLP
SQL
Git

Run Screening

JD Skills:

```
[
0 : "sql"
1 : "machine learning"
2 : "tensorflow"
3 : "nlp"
4 : "deep learning"
5 : "git"
6 : "python"
]
```

Ranking Completed Successfully!

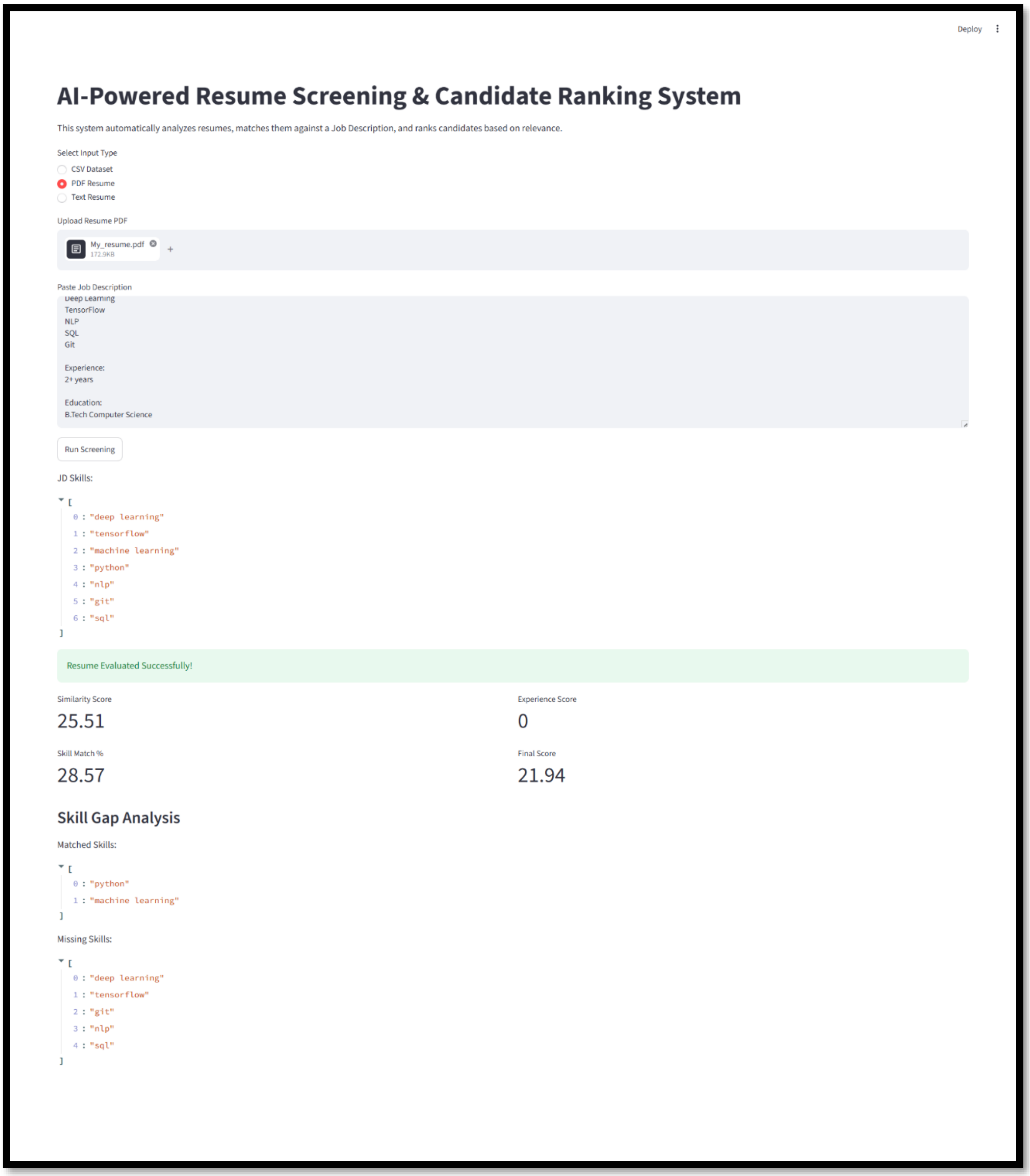
Top Ranked Candidates

Top Candidate Skills

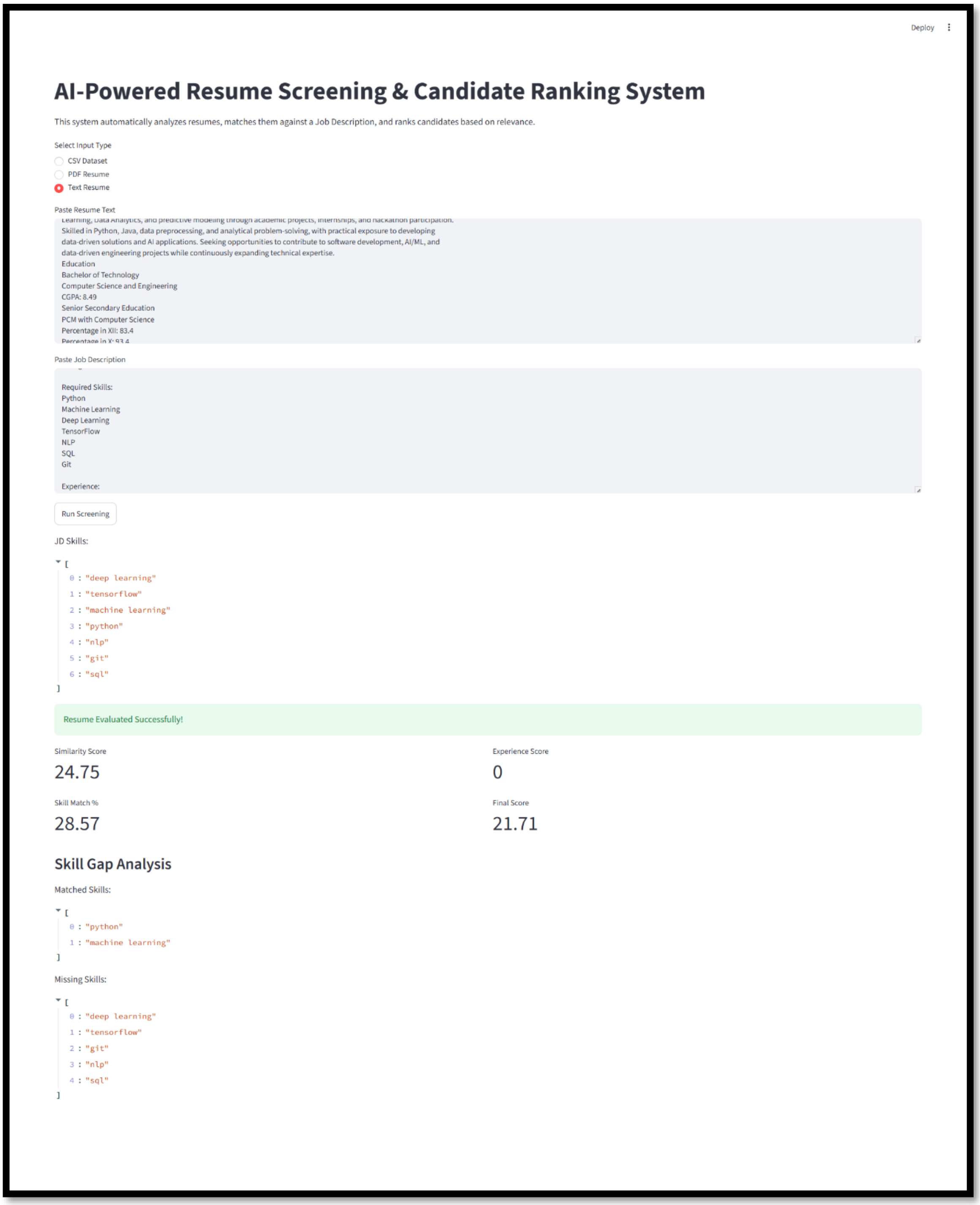
	Category	skills	skill_match_score	similarity_score	final_score
0	AGRICULTURE	javascript html sql python tensorflow c mongodb flask css data visualizati	42.8571	100	71.4286
1	ENGINEERING	data visualization html sql python machine learning c data analysis statistics	42.8571	85.8545	67.1849
2	ENGINEERING	sql postgresql mongodb linux git python	42.8571	67.7202	61.7446
3	AUTOMOBILE	sql machine learning java c excel data analysis statistics python	42.8571	61.7613	59.957
4	CONSULTANT	javascript data visualization sql machine learning java c azure data science d	42.8571	61.1547	59.775
5	BANKING	javascript html sql aws java nlp kubernetes c nodejs deep learning linu	57.1429	34.9142	59.0457
6	AVIATION	javascript html sql c excel postgresql css linux mysql docker git pyth	42.8571	50.3114	56.522
7	INFORMATION-TECHNOLOGY	sql excel python	28.5714	58.6328	51.8756
8	ENGINEERING	javascript html sql aws java c mongodb github css python	28.5714	56.1456	51.1294
9	ENGINEERING	c sql tableau python	28.5714	50.4086	49.4083

	Rank	Category	similarity_score	skill_match_score	experience_score	final_score
0	1	AGRICULTURE	100	42.8571	100	71.4286
1	2	ENGINEERING	85.8545	42.8571	100	67.1849
2	3	ENGINEERING	67.7202	42.8571	100	61.7446
3	4	AUTOMOBILE	61.7613	42.8571	100	59.957
4	5	CONSULTANT	61.1547	42.8571	100	59.775
5	6	BANKING	34.9142	57.1429	100	59.0457
6	7	AVIATION	50.3114	42.8571	100	56.522
7	8	INFORMATION-TECHNOLOGY	58.6328	28.5714	100	51.8756
8	9	ENGINEERING	56.1456	28.5714	100	51.1294
9	10	ENGINEERING	50.4086	28.5714	100	49.4083

Download Results CSV



Deploy
AI-Powered Resume Screening & Candidate Ranking System
This system automatically analyzes resumes, matches them against a Job Description, and ranks candidates based on relevance.
Select Input Type
CSV Dataset
PDF Resume
Text Resume
Upload Resume PDF
My_resume.pdf
172.9KB
Paste Job Description
Deep Learning
TensorFlow
NLP
SQL
Git
Experience:
2+ years
Education:
B.Tech Computer Science
Run Screening
JD Skills:
[
0 : "deep learning"
1 : "tensorflow"
2 : "machine learning"
3 : "python"
4 : "nlp"
5 : "git"
6 : "sql"
]
Resume Evaluated Successfully!
Similarity Score
25.51
Experience Score
0
Skill Match %
28.57
Final Score
21.94
Skill Gap Analysis
Matched Skills:
[
0 : "python"
1 : "machine learning"
]
Missing Skills:
[
0 : "deep learning"
1 : "tensorflow"
2 : "git"
3 : "nlp"
4 : "sql"
]



The ranking mechanism effectively prioritized resumes with higher relevance to the provided job description.

9. Limitations

Despite achieving the project objectives, several limitations remain:

- Skill extraction relies on predefined dictionaries
- Experience extraction depends on textual patterns
- TF-IDF does not capture deep semantic meaning
- Dataset category labels may contain inconsistencies
- Contextual understanding is limited compared to modern LLMs

10. Future Enhancements

Potential future improvements include:

- BERT-based semantic matching
- Sentence Transformer embeddings
- Resume summarization using LLMs
- Named Entity Recognition (NER)
- Recruiter feedback integration
- Bias detection and fairness evaluation
- Multi-job comparison support

11. Conclusion

The AI-Powered Resume Screening and Candidate Ranking System successfully demonstrates the application of NLP and Machine Learning techniques in recruitment automation.

The system supports multiple resume formats, performs automated candidate evaluation, and provides recruiter-friendly ranking outputs through a Streamlit dashboard. The project represents a practical implementation of ATS-inspired technology and serves as a strong foundation for future enhancements involving advanced NLP and Large Language Models.